

ABHINAV VERMA

Hardware & PCB Design Engineer | Embedded Systems | RF/Mixed-Signal | FPGA Design

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[My Portfolio](#)

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PROFESSIONAL EXPERIENCE

Hardware & PCB Design Engineer · Applied Electro Magnetix Pvt. Ltd. (AEM)

Jan 2026 – Present

Clients: Bharat Electronics Limited (BEL) · Velmenni

- Designed and debugged multi-layer PCB hardware for a radar signal processing system (BEL ATC), achieving zero critical failures during validation.
- Led hardware bring-up and testing for a gimbal aligner system (Velmenni), improving alignment precision by 40% for stable optical communication.

Electronic Design Engineer · Satyam Software Solutions Pvt. Ltd. (S-CUBE)

Apr 2023 – Jan 2026

Clients: Airtel Payment Bank · NPCI · DMRC · CMRL

- Designed and certified an Open-Loop Card Reader with Android application board, achieving full EMVCo L1, L2, and L3 compliance — enabling secure, zero-defect payment processing across 3 certification tiers.
- Led end-to-end PCB layout and hardware architecture for a Card Dispenser module (single/dual stacker) deployed for Airtel Payment Bank in partnership with NPCI, reducing integration issues by 30% through DFM-first design.
- Engineered a 4-interface USB-HUB (GPIOs + UART + HDMI) for DMRC Distributed TVMs, enabling reliable peripheral interfacing across 5+ metro station deployments.
- Developed a DC UPS system for Chennai Metro (CMRL), delivering uninterrupted power backup for TVMs and AFC gates across 10+ installations, with UART-controlled Auto/Manual power-off functionality.
- Reduced PCB layout errors by 25% through reverse engineering of the DMRC DC Booster Module and redesigning Platform Screen Door and Flap-Controller PCBs, resolving 8+ pre-existing design defects.
- Maintained structured project documentation across 5+ concurrent hardware projects, ensuring 100% compliance with organizational SOPs and client handover standards.

Engineering Intern · Aerial Delivery R&D Establishment (DRDO)

Jun 2022 – Jul 2022

- Built a Python-based radar application to compute GPS coordinate distances and perform real-time location tracking, reducing manual calculation effort by 80% for defense navigation testing workflows.

TECHNICAL SKILLS

EDA Tools: Altium Designer, Allegro, KiCad, SOLIDWORKS

Hardware Design: Schematic Capture, Multi-layer PCB Layout, Component Selection, DFM/DFT, Signal Integrity, Power Integrity, Gerber Generation, BOM Management, Hardware Design Review, Bring-up & Debug

Domains: Analog, Digital, RF/Mixed-Signal, FPGA, EMI/EMC Compliance, Power Electronics, Embedded Systems

Protocols: CAN, Ethernet, I²C, I²S, PCIe, SPI, UART, USB, JTAG

Programming & Simulation: C, C++, Python, Verilog HDL, LTspice, RTL

Test & Measurement: Oscilloscope, Multimeter, LCR Meter, Bench Power Supply, Electronic Load, Soldering Station

ACADEMIC PROJECTS

Wireless Digital Notice Board — LoRa-based Mobile-Controlled Communication System

Jun 2023

- Designed a long-range LoRa-based digital notice board with real-time wireless message updates via mobile app.

LOS Propagation Model — Wireless Optical Communication Data Transmission System

Apr 2020

- Developed an LOS optical wireless communication system using Arduino, LED, and photodiode with error-correction for reliable data transmission.

EDUCATION

B.Tech. in Electronics & Communication Engineering

Aug 2019 – May 2023

Dr. A.P.J. Abdul Kalam Technical University · CGPA: 7.68 / 10

CERTIFICATIONS

- ADRDE Training Certificate — Radar-based Distance Measurement & GPS Location Tracking | DRDO, 2022
- Programming with C/C++ | Internshala, 2022
- Verilog HDL: VLSI Hardware Design Comprehensive | Udemy, 2023
- Circuit Design Training Certification | S-CUBE, 2023